

ASSESSMENT – 1

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Register number :23MISO382

Branch: MIS

Slot: L3+L4

Subject: Agile development process and devops lab

Faculty: Dr . Sweta Bhattacharya

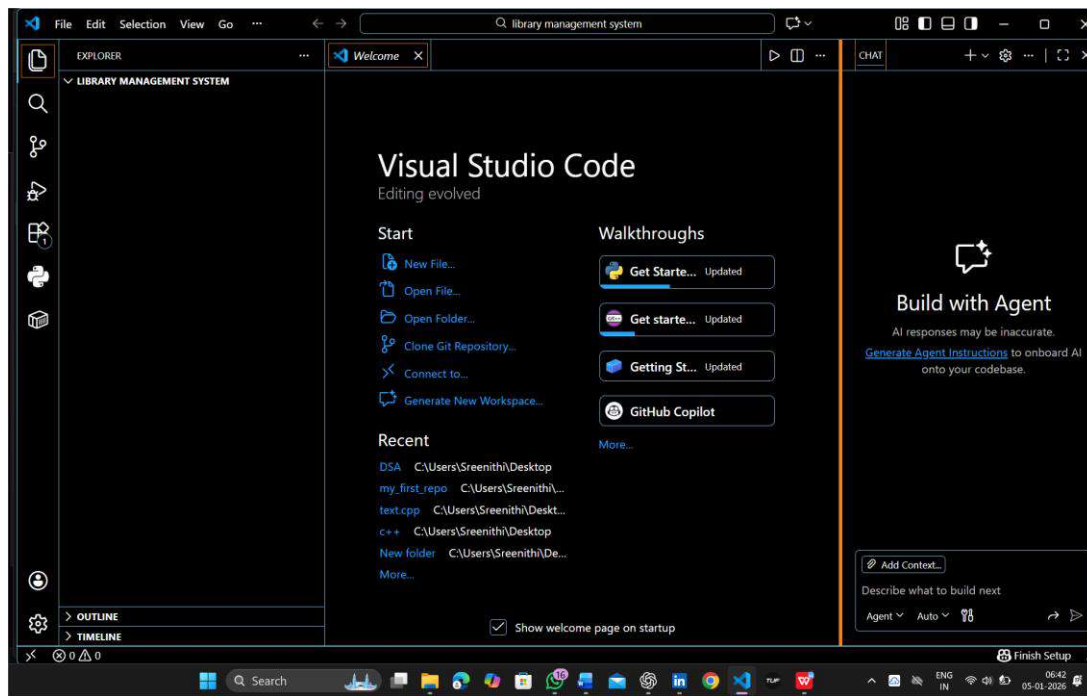
Use appropriate Git Commands to Execute the following case studies in VS Code

Case Study 1: New Project Version Control Setup

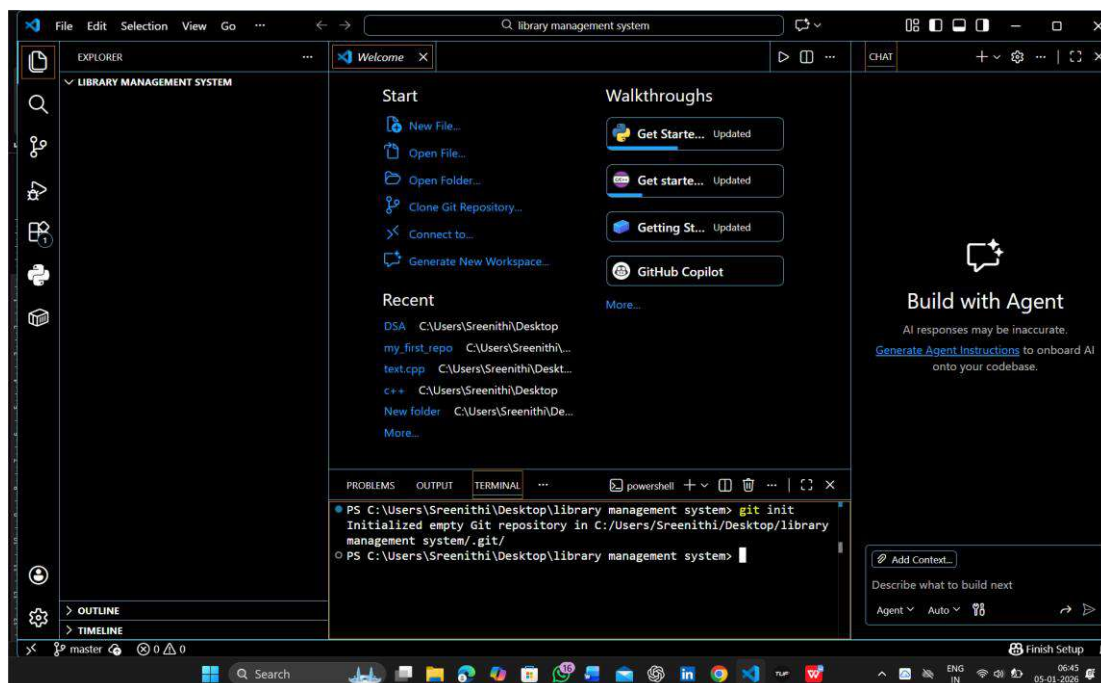
Scenario: You are assigned to develop a Library Management System. Before coding, your team decides to enable version control.

Tasks:

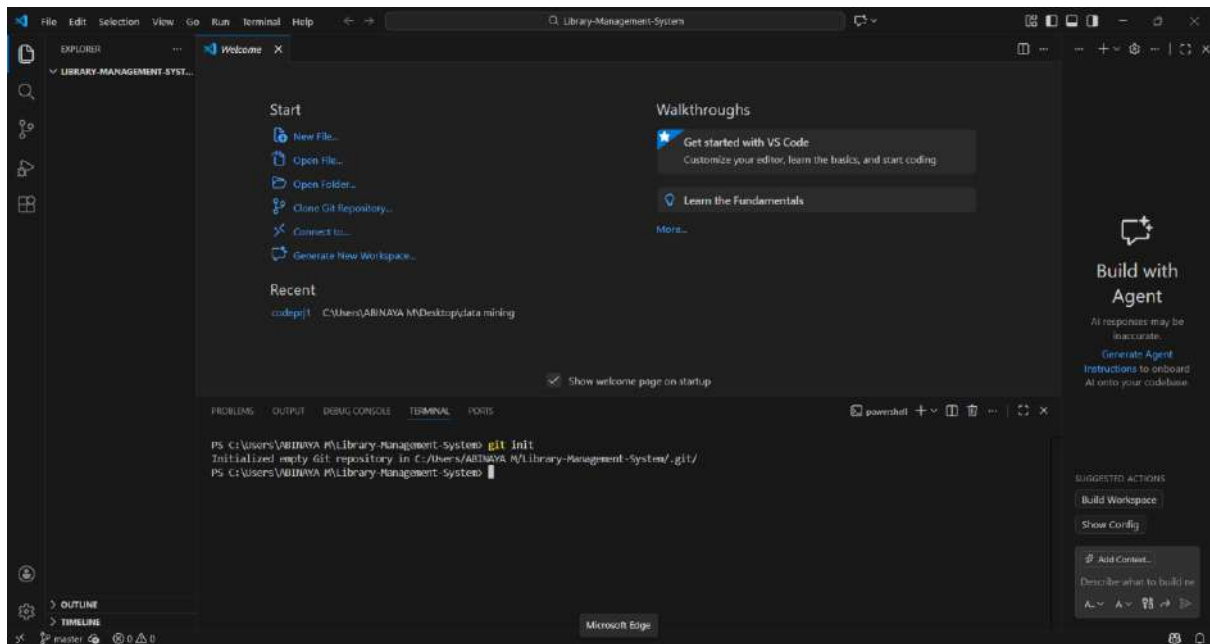
- Create a project folder and open it in VS Code



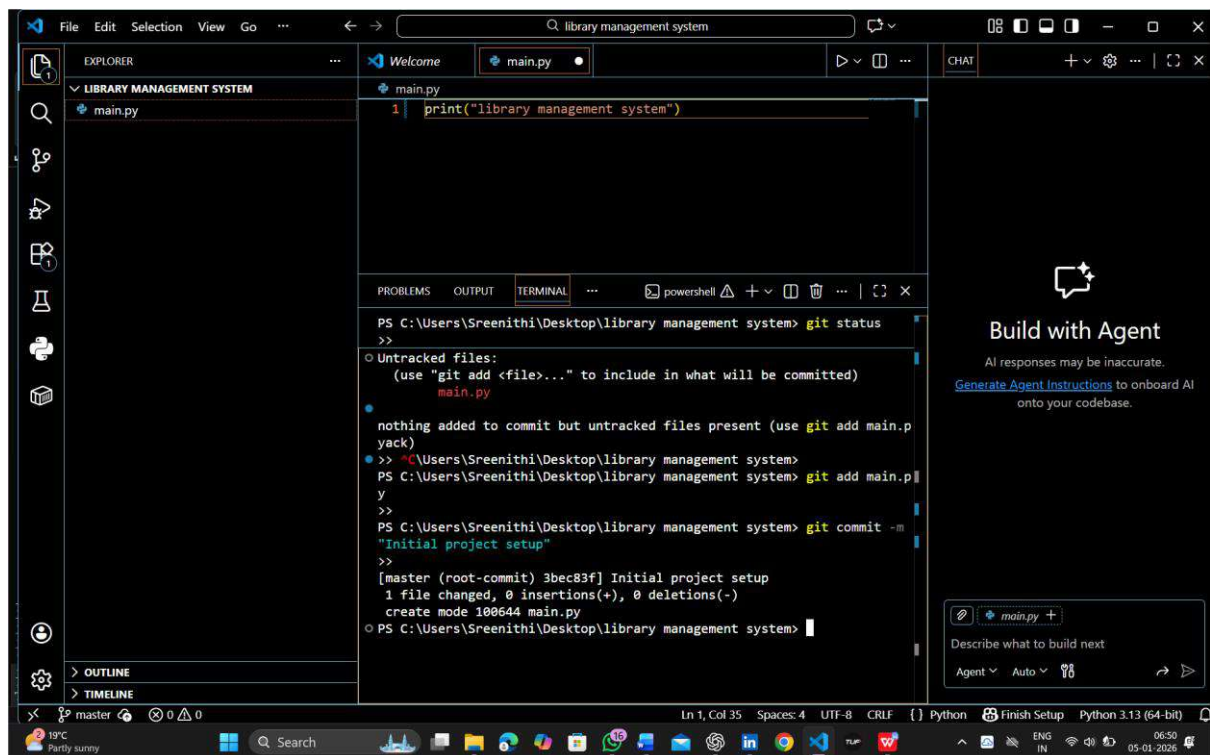
- Initialize a Git repository



- Create a Python file and track it using Git



- Commit the initial version

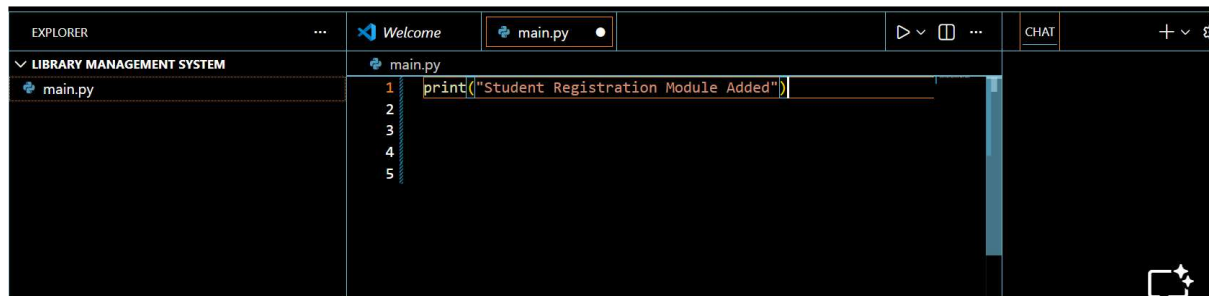


Case Study 2: Tracking Daily Code Changes

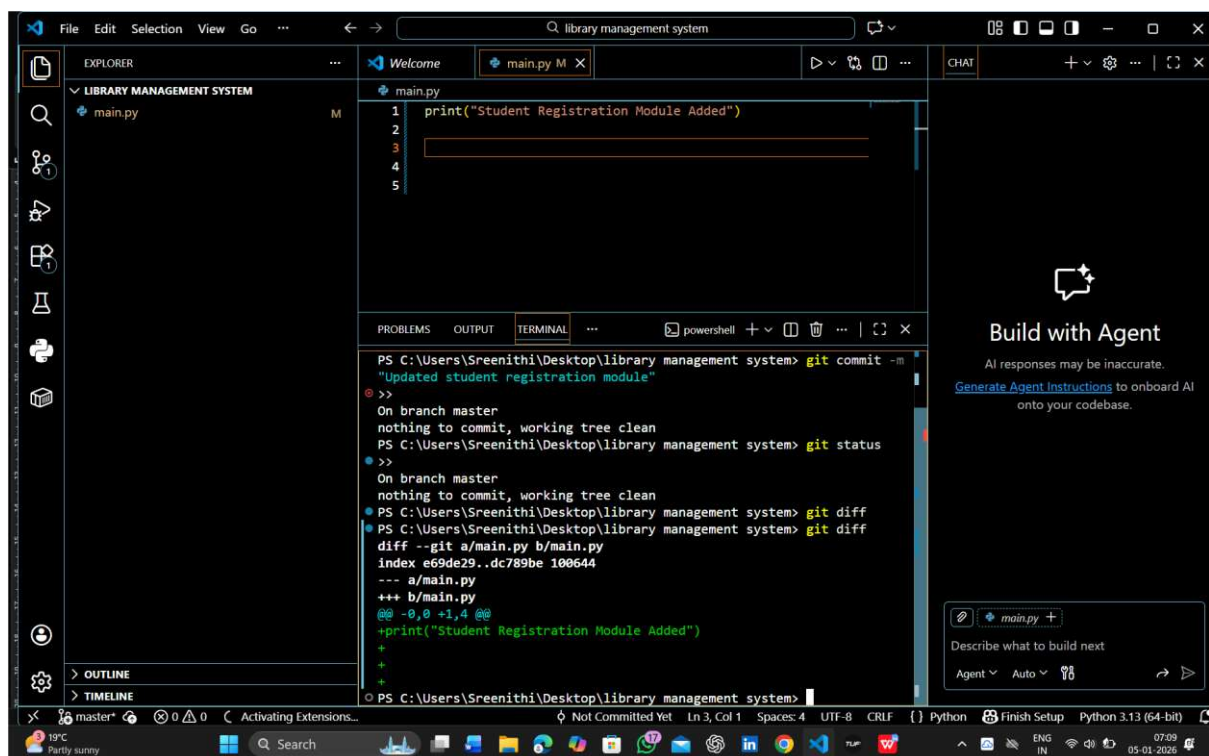
Scenario: You are working on a Student Registration Module. You make small updates daily and need to record them.

Tasks:

Modify an existing file



- Check file differences



-stage and commit changes

```
File Edit Selection View Go ... library management system
EXPLORER
LIBRARY MANAGEMENT SYSTEM
main.py
main.py
1 print("Student Registration Module Added")
2
3
4
5
TERMINAL
PS C:\Users\Sreenithi\Desktop\library management system> git commit -m "Updated student registration module"
>>
PS C:\Users\Sreenithi\Desktop\library management system> git status
>>
On branch master
nothing to commit, working tree clean
PS C:\Users\Sreenithi\Desktop\library management system> git diff
PS C:\Users\Sreenithi\Desktop\library management system> git diff
diff --git a/main.py b/main.py
index e69de29..dc789be 100644
--- a/main.py
+++ b/main.py
@@ -0,0 +1,4 @@
+print("Student Registration Module Added")
+
+
+
PS C:\Users\Sreenithi\Desktop\library management system> git add main.py
y
>>
Not Committed Yet Ln 3, Col 1 Spaces: 4 UTF-8 CRLF Python Finish Setup Python 3.13 (64-bit)
19°C Mostly sunny
```

- View commit history

```
File Edit Selection View Go ... library management system
EXPLORER
LIBRARY MANAGEMENT SYSTEM
main.py
main.py
1 print("Student Registration Module Added")
2
3
4
5
TERMINAL
PS C:\Users\Sreenithi\Desktop\library management system> git diff
@@ -0,0 +1,4 @@
+print("Student Registration Module Added")
+
+
+
PS C:\Users\Sreenithi\Desktop\library management system> git add main.py
y
>> git commit -m "Updated student registration module"
>>
[master 16c2c15] Updated student registration module
1 file changed, 4 insertions(+)
PS C:\Users\Sreenithi\Desktop\library management system> git log
>>
commit 16c2c1514a68f0236ad08e1c97ebdd3ecffb37eb (HEAD -> master)
Author: sreenithi07 <sreenithi.k07@gmail.com>
Date: Mon Jan 5 07:11:42 2026 +0530

    Updated student registration module

commit 3bec83ff19ea6db2d3fa59b364f9c9ae35f37376
Author: sreenithi07 <sreenithi.k07@gmail.com>
Date: Mon Jan 5 06:50:36 2026 +0530

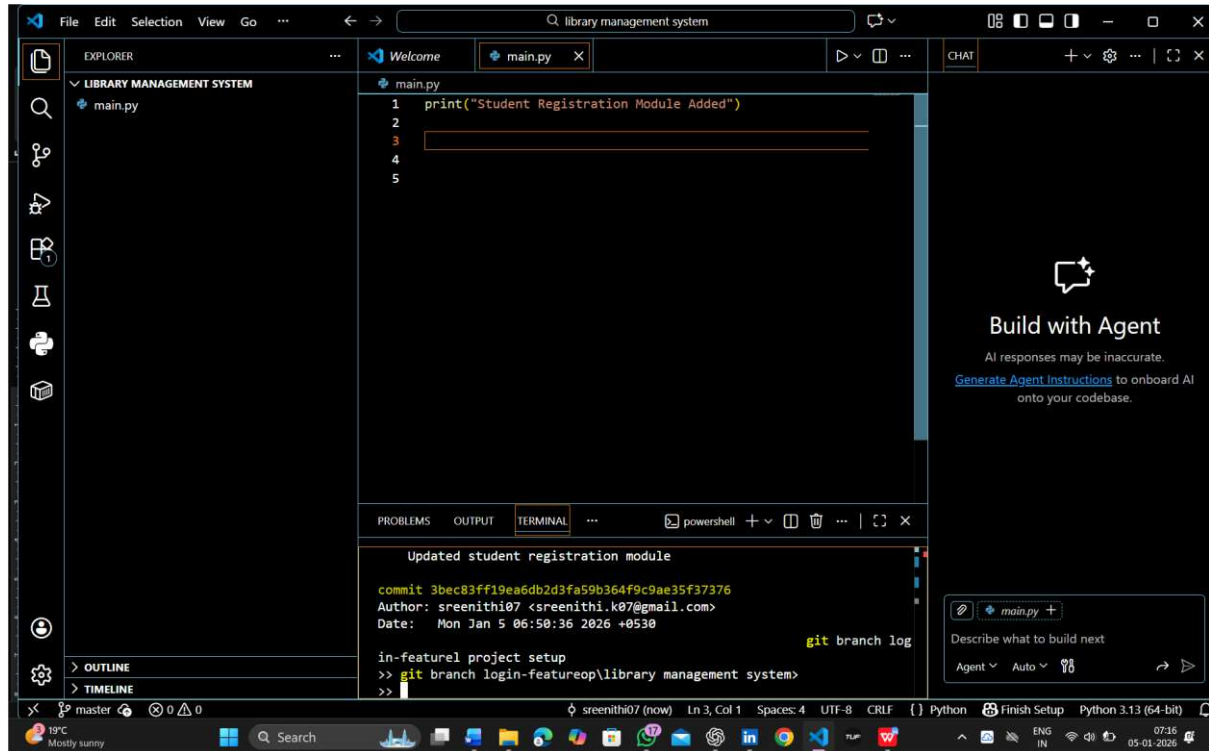
    Initial project setup
PS C:\Users\Sreenithi\Desktop\library management system>
sreenithi07 (now) Ln 3, Col 1 Spaces: 4 UTF-8 CRLF Python Finish Setup Python 3.13 (64-bit)
19°C Mostly sunny
```

Case Study 3: Feature Development Using Branches

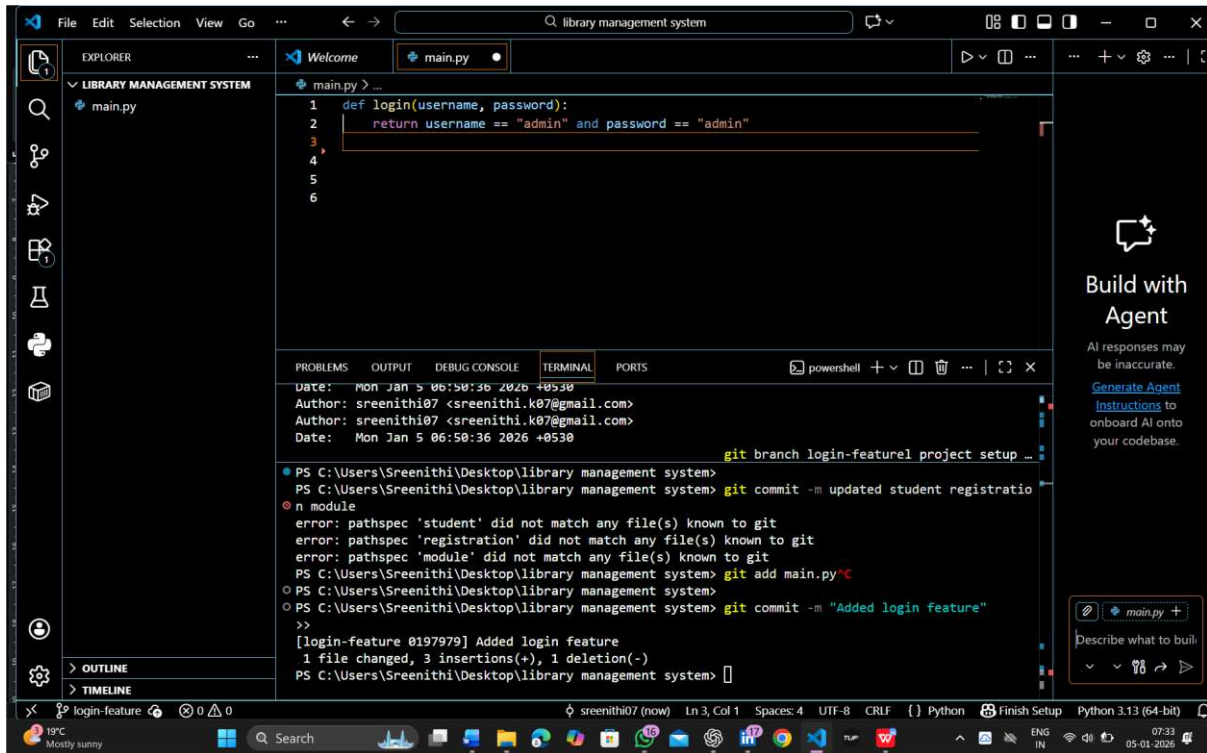
Scenario: A new Login feature must be developed without disturbing main code.

Tasks:

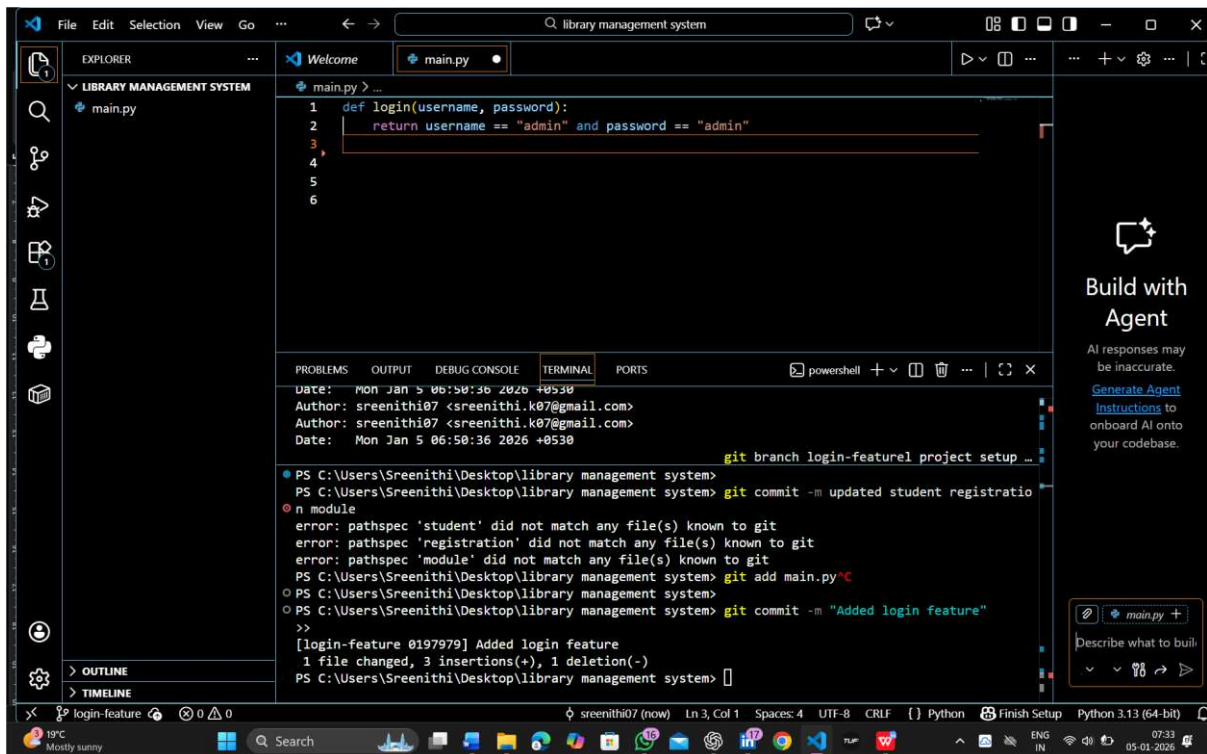
- Create a new branch



- Add login-related code



- Commit changes



- Switch between branches

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Author: sreenithi07 <sreenithi.k07@gmail.com>
Author: sreenithi07 <sreenithi.k07@gmail.com>
Date: Mon Jan 5 06:50:36 2026 +0530

git branch login-feature1 project setup ...

PS C:\Users\Sreenithi\Desktop\library management system>
PS C:\Users\Sreenithi\Desktop\library management system> git commit -m updated student registrati
n module
error: pathspec 'student' did not match any file(s) known to git
error: pathspec 'registration' did not match any file(s) known to git
error: pathspec 'module' did not match any file(s) known to git
PS C:\Users\Sreenithi\Desktop\library management system> git add main.py^C
PS C:\Users\Sreenithi\Desktop\library management system>
PS C:\Users\Sreenithi\Desktop\library management system> git commit -m "Added login feature"
>>
[login-feature 0197979] Added login feature
1 file changed, 3 insertions(+), 1 deletion(-)
PS C:\Users\Sreenithi\Desktop\library management system> git checkout main
error: pathspec 'main' did not match any file(s) known to git
PS C:\Users\Sreenithi\Desktop\library management system>
```

Case Study 4: Merging Feature into Main Code

Scenario: The login feature is tested and approved.

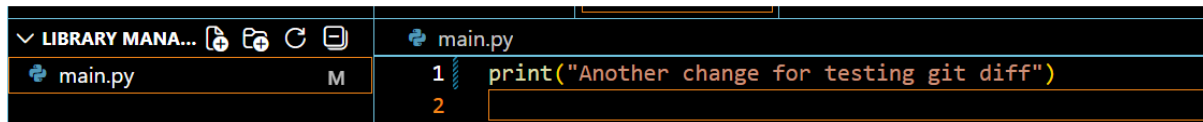
Tasks:

- Switch to main branch
- Merge feature branch

```
File Edit Selection View Go ... library management system
EXPLORER
LIBRARY MANAGEMENT SYSTEM
main.py M
main.py
1 print("Another change for testing git diff")
2
3

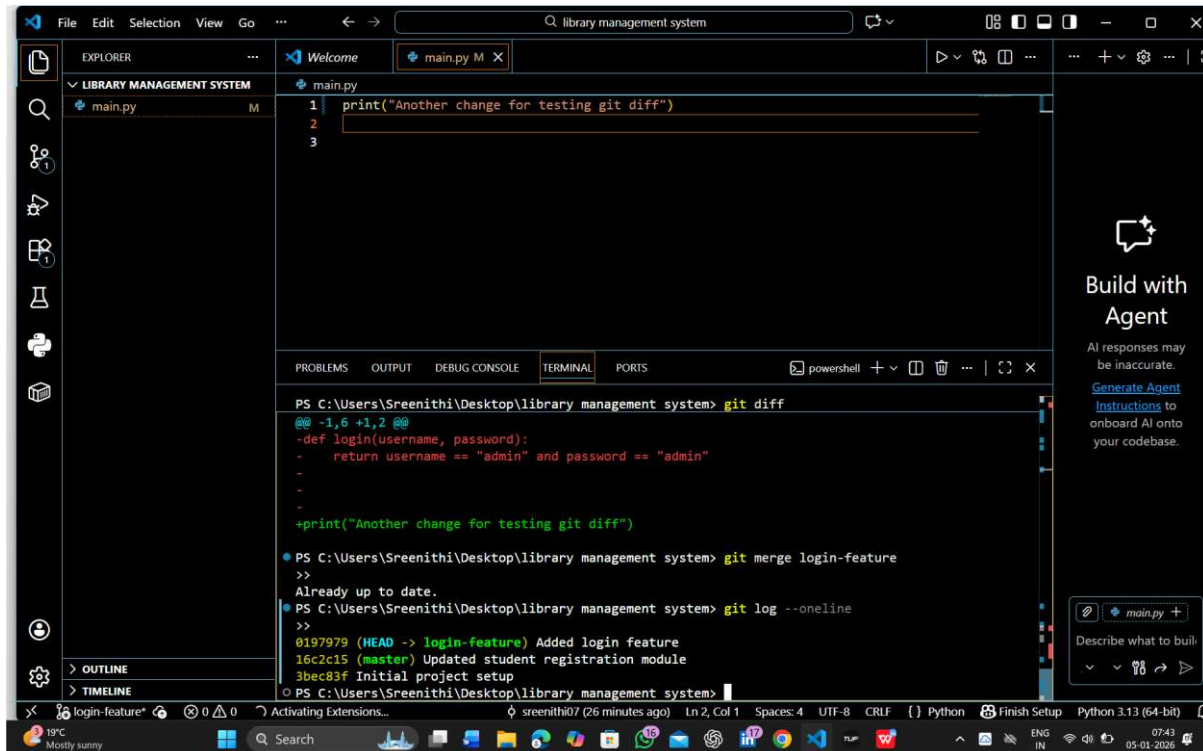
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Sreenithi\Desktop\library management system> git checkout main
PS C:\Users\Sreenithi\Desktop\library management system> git diff
diff --git a/main.py b/main.py
index 8dee0e0..2ab99d4 100644
--- a/main.py
+++ b/main.py
@@ -1,6 +1,2 @@
-def login(username, password):
-    return username == "admin" and password == "admin"
-
-
-+print("Another change for testing git diff")
PS C:\Users\Sreenithi\Desktop\library management system> git merge login-feature
>>
Already up to date.
PS C:\Users\Sreenithi\Desktop\library management system>
```

-verify merged code:



```
main.py
1 print("Another change for testing git diff")
2
```

-check commit history:



```
PS C:\Users\Sreenithi\Desktop\library management system> git diff
@@ -1,6 +1,2 @@
-def login(username, password):
-    return username == "admin" and password == "admin"
-
-
-+print("Another change for testing git diff")

PS C:\Users\Sreenithi\Desktop\library management system> git merge login-feature
>>
Already up to date.

PS C:\Users\Sreenithi\Desktop\library management system> git log --oneline
>>
0197979 (HEAD -> login-feature) Added login feature
16c2c15 (master) Updated student registration module
3bec83f Initial project setup

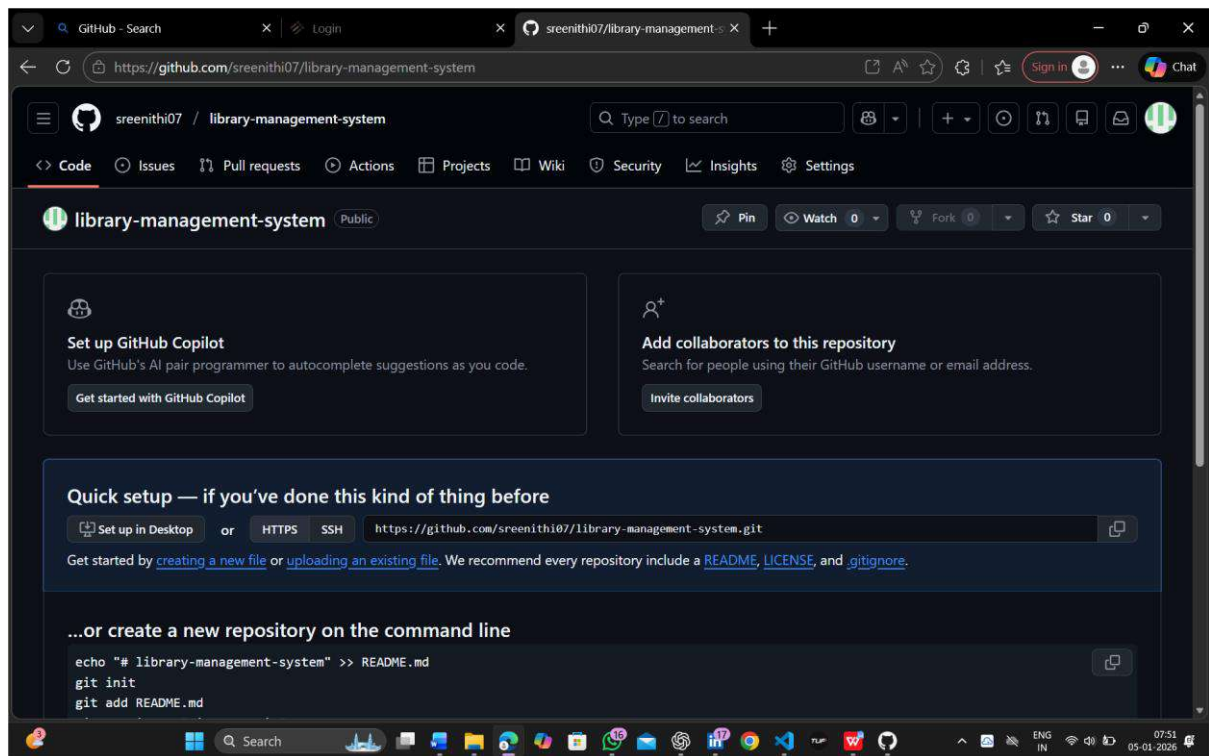
PS C:\Users\Sreenithi\Desktop\library management system>
```

Case Study 5: Working with Remote Repository

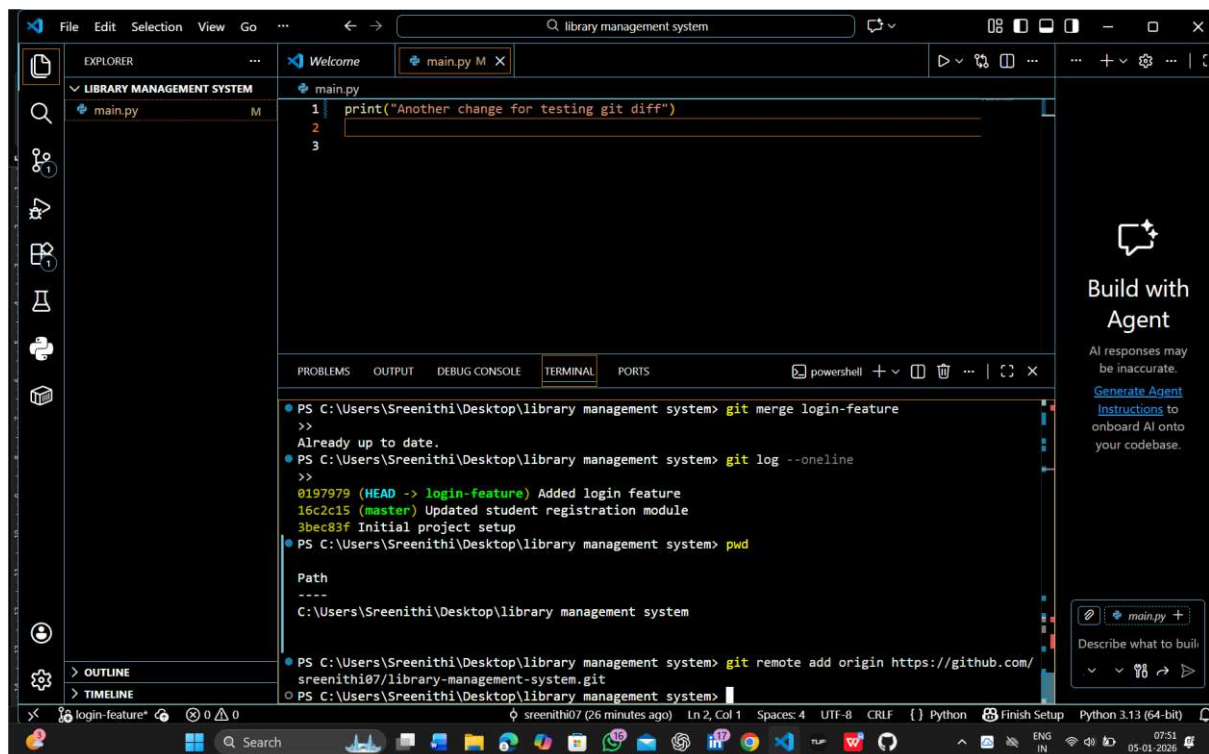
Scenario: The project must be shared using GitHub

Tasks:

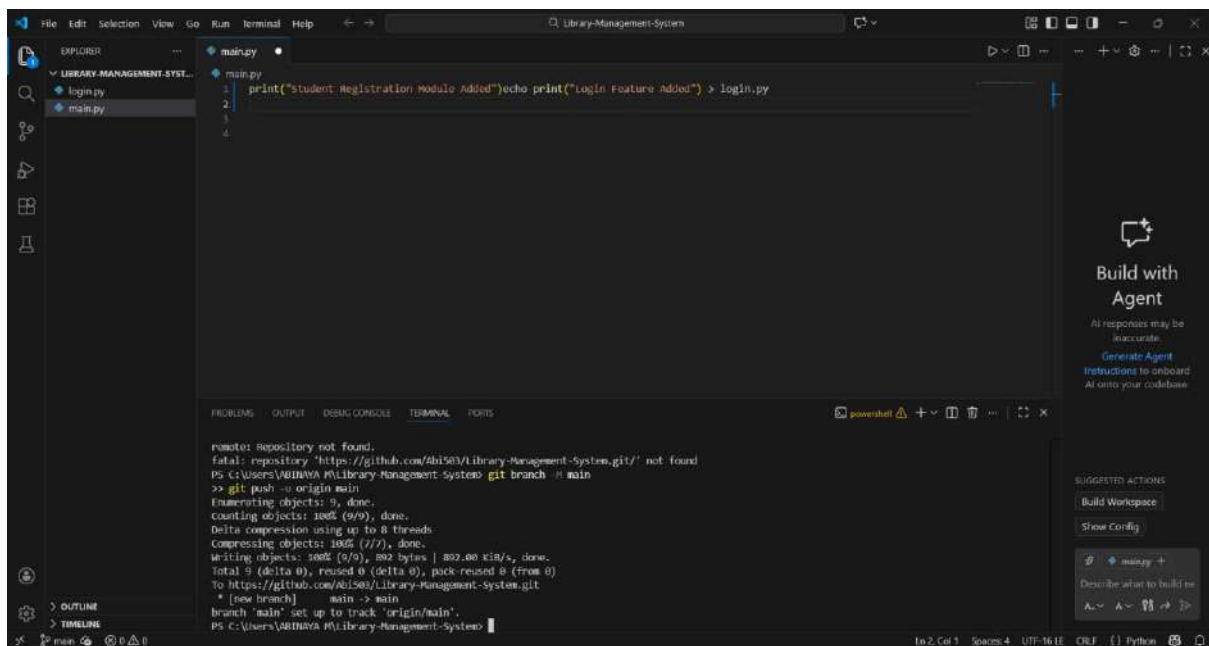
Create remote repository



-add remote origin:



- Push code to GitHub



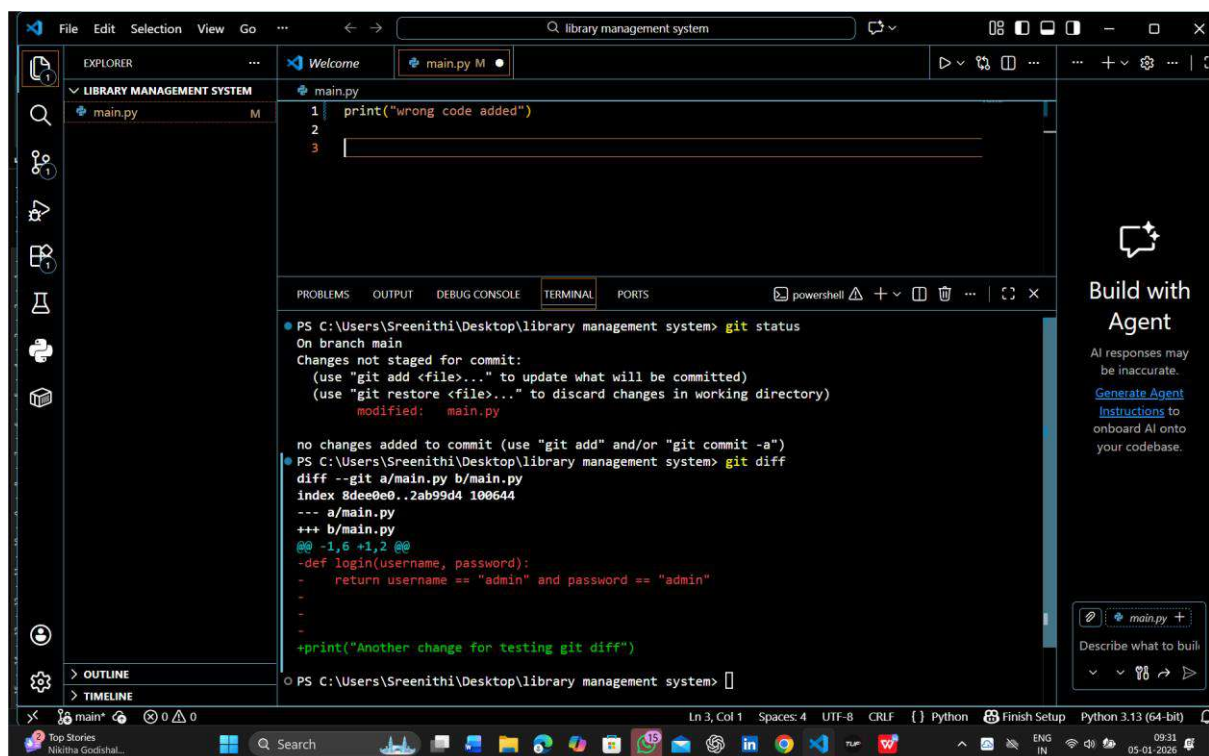
Use case 6:

Undoing mistakes

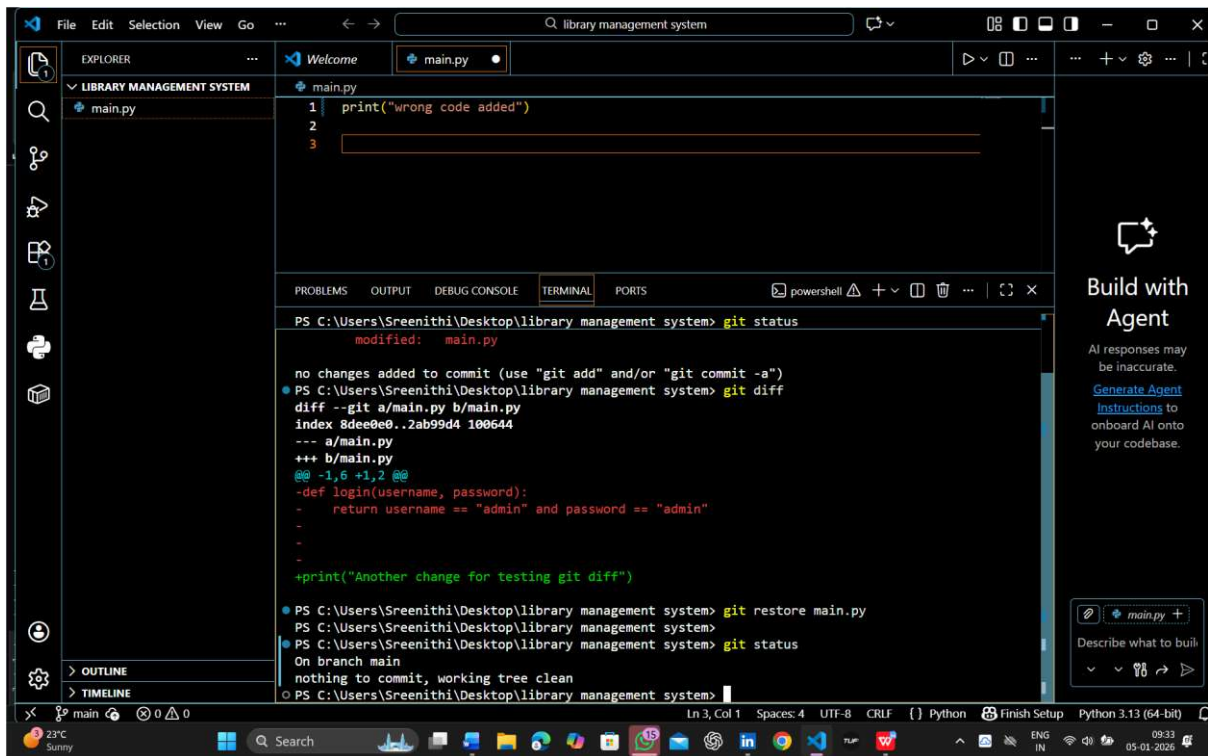
Scenario: a file is modified incorrectly.

Tasks:

- View file changes



- Discard uncommitted changes



```
1 print("wrong code added")
2
3

PS C:\Users\Sreenithi\Desktop\library management system> git status
modified:   main.py

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\Sreenithi\Desktop\library management system> git diff
diff --git a/main.py b/main.py
index 8dee0e0..2ab99d4 100644
--- a/main.py
+++ b/main.py
@@ -1,6 +1,2 @@
-def login(username, password):
-    return username == "admin" and password == "admin"
-
-
-+print("Another change for testing git diff")

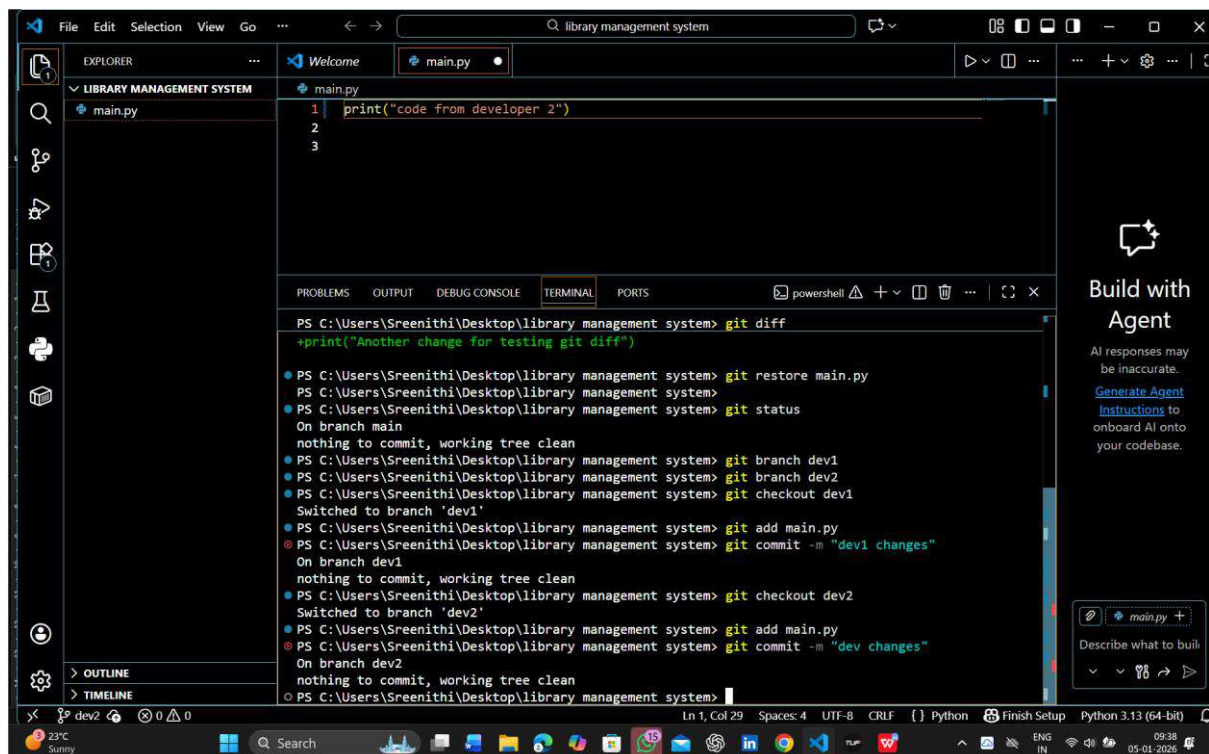
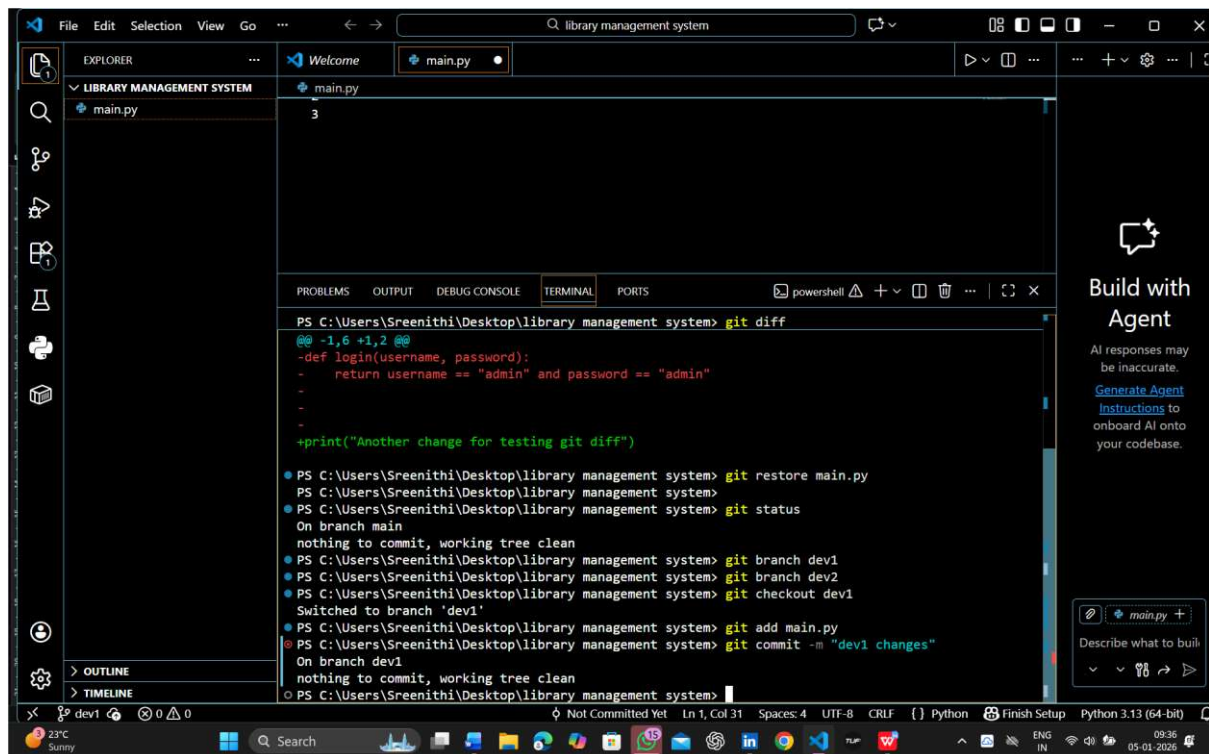
PS C:\Users\Sreenithi\Desktop\library management system> git restore main.py
PS C:\Users\Sreenithi\Desktop\library management system> git status
On branch main
nothing to commit, working tree clean
PS C:\Users\Sreenithi\Desktop\library management system>
```

Case Study 7: Handling Merge Conflicts

Scenario: Two developers edited the same file.

Tasks:

- Merge branches
- Resolve conflicts
- Commit resolved changes



Merging 1 and 2 and conflict resolved successfully:

The screenshot shows the Visual Studio Code interface. The Explorer sidebar on the left shows a project named 'LIBRARY MANAGEMENT SYSTEM' with a file 'main.py'. The main editor area shows the content of 'main.py', which contains three lines of code: `1 print("code from developer 2")`, `2`, and `3`. The Terminal panel at the bottom displays a series of git commands and their outputs, including creating branches 'dev1' and 'dev2', adding 'main.py', committing changes, and merging 'dev1' into 'dev2'. The status bar at the bottom indicates the current branch is 'dev2'. On the right side, there is a sidebar with the heading 'Build with Agent' and a button to 'Generate Agent'.

This screenshot shows a different view of the Visual Studio Code interface. The Explorer sidebar on the left shows the project 'LIBRARY-MANAGEMENT-SYST...' with two files: 'login.py' and 'main.py'. The main editor area shows the content of 'main.py', which contains four lines of code: `1 print("Master + Login feature merged")`, `2`, `3`, and `4`. The status bar at the bottom indicates the current branch is 'dev2'.

Case Study 8: Ignoring Unwanted Files

Scenario: Log and temporary files should not be tracked.

Tasks:

- Create .gitignore file:

The screenshot shows the Visual Studio Code interface with a terminal window open. The Explorer pane on the left shows a project named 'LIBRARY MANAGEMENT SYSTEM' with files '.gitignore', 'echo', and 'main.py'. The terminal window displays the following commands and output:

```
PS C:\Users\Sreenithi\Desktop\library management system> *.log
At line:1 char:1
+ *.log
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (*.log:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Sreenithi\Desktop\library management system> touch .gitignore
touch : The term 'touch' is not recognized as the name of a cmdlet, function, script file, or
operable program. Check the spelling of the name, or if a path was included, verify that the path
is correct and try again.
At line:1 char:1
+ touch .gitignore
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (touch:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Sreenithi\Desktop\library management system> echo > .gitignore

cmdlet Write-Output at command pipeline position 1
Supply values for the following parameters:
InputObject[0]:
PS C:\Users\Sreenithi\Desktop\library management system>
```

On the right side of the terminal, there is a 'Build with Agent' panel with a search bar and a 'Describe what to build' button.

- Add ignore patterns

The screenshot shows the Visual Studio Code interface with a terminal window open. The Explorer pane on the left shows a project named 'LIBRARY MANAGEMENT SYSTEM' with files '.gitignore', 'echo', and 'main.py'. The terminal window displays the following commands and output:

```
PS C:\Users\Sreenithi\Desktop\library management system> *.log
>> *.tmp
>> __pycache__/_
__pycache__/_ : The term '__pycache__/_' is not recognized as the name of a cmdlet, function,
script file, or operable program. Check the spelling of the name, or if a path was included,
verify that the path is correct and try again.
At line:3 char:1
+ __pycache__/_
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (__pycache__/_:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Sreenithi\Desktop\library management system> git add .gitignore
git: 'add.gitignore' is not a git command. See 'git --help'.
PS C:\Users\Sreenithi\Desktop\library management system> git commit -m "added gitignore file"
On branch dev2
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        echo
nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\Sreenithi\Desktop\library management system>
```

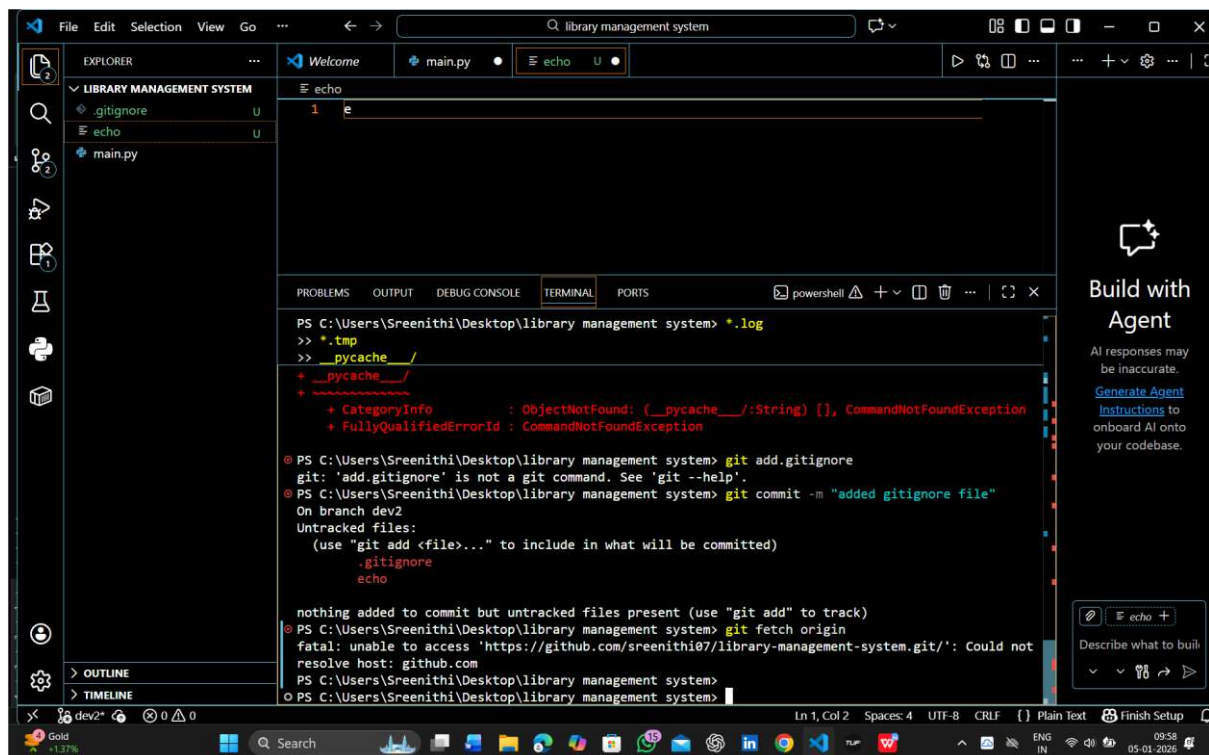
On the right side of the terminal, there is a 'Build with Agent' panel with a search bar and a 'Describe what to build' button.

Case Study 9: Collaborating with Team

Scenario: A teammate pushed new changes.

Tasks:

- Fetch latest changes



The screenshot shows the Visual Studio Code interface with a terminal window open. The terminal displays the following commands and output:

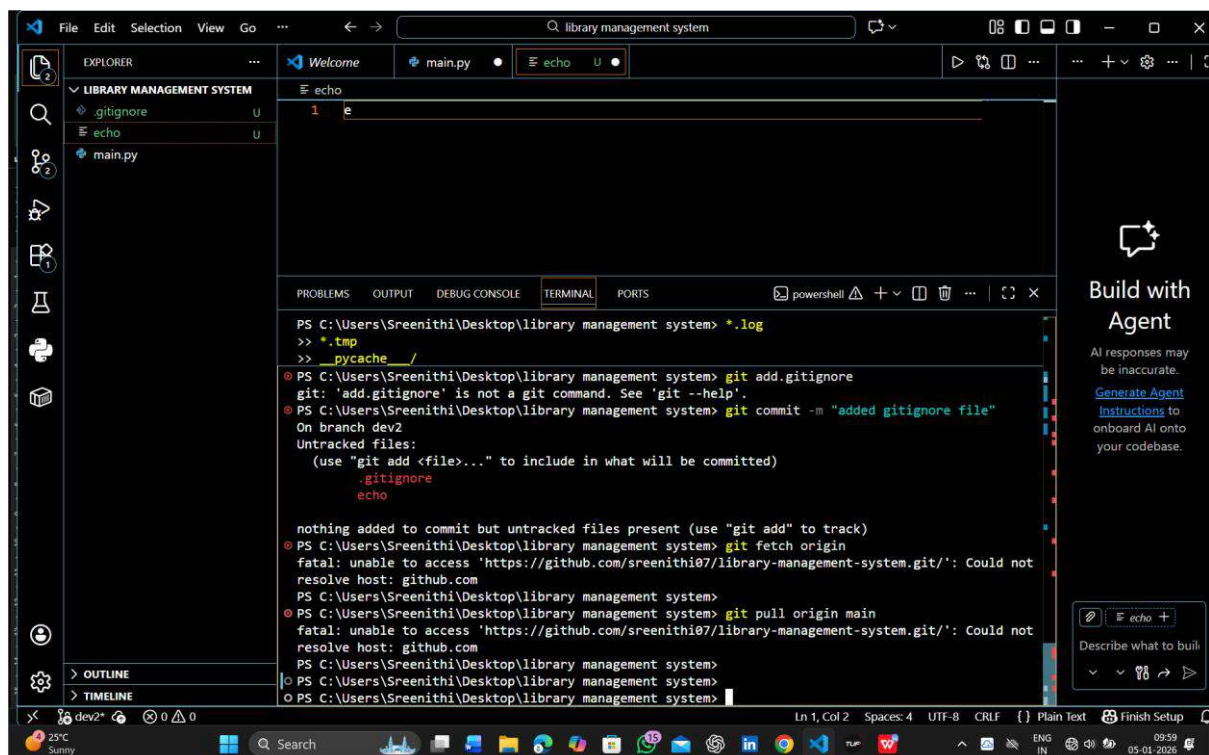
```
PS C:\Users\Sreenithi\Desktop\library management system> *.log
>> *.tmp
>> _pycache_/
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (._pycache_/:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Sreenithi\Desktop\library management system> git add.gitignore
git: 'add.gitignore' is not a git command. See 'git --help'.
PS C:\Users\Sreenithi\Desktop\library management system> git commit -m "added gitignore file"
On branch dev2
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        echo

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\Sreenithi\Desktop\library management system> git fetch origin
fatal: unable to access 'https://github.com/sreenithi07/library-management-system.git/': Could not
resolve host: github.com
PS C:\Users\Sreenithi\Desktop\library management system>
```

The Explorer view on the left shows the file structure of the 'LIBRARY MANAGEMENT SYSTEM' project, including .gitignore, echo, and main.py. The bottom status bar indicates the current branch is 'dev2'.

- Pull updates



The screenshot shows the Visual Studio Code interface with a terminal window open. The terminal displays the following commands and output:

```
PS C:\Users\Sreenithi\Desktop\library management system> *.log
>> *.tmp
>> _pycache_/
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (._pycache_/:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Sreenithi\Desktop\library management system> git add.gitignore
git: 'add.gitignore' is not a git command. See 'git --help'.
PS C:\Users\Sreenithi\Desktop\library management system> git commit -m "added gitignore file"
On branch dev2
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        echo

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\Sreenithi\Desktop\library management system> git fetch origin
fatal: unable to access 'https://github.com/sreenithi07/library-management-system.git/': Could not
resolve host: github.com
PS C:\Users\Sreenithi\Desktop\library management system> git pull origin main
fatal: unable to access 'https://github.com/sreenithi07/library-management-system.git/': Could not
resolve host: github.com
PS C:\Users\Sreenithi\Desktop\library management system>
```

The Explorer view on the left shows the file structure of the 'LIBRARY MANAGEMENT SYSTEM' project, including .gitignore, echo, and main.py. The bottom status bar indicates the current branch is 'dev2'.

```
PS C:\Users\Sreenithi\Desktop\library management system> git fetch origin
fatal: unable to access 'https://github.com/sreenithi07/library-management-system.git/': Could not
resolve host: github.com
PS C:\Users\Sreenithi\Desktop\library management system> git add main.py
PS C:\Users\Sreenithi\Desktop\library management system> git commit -m "resolved pull conflict"
On branch dev2
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        echo

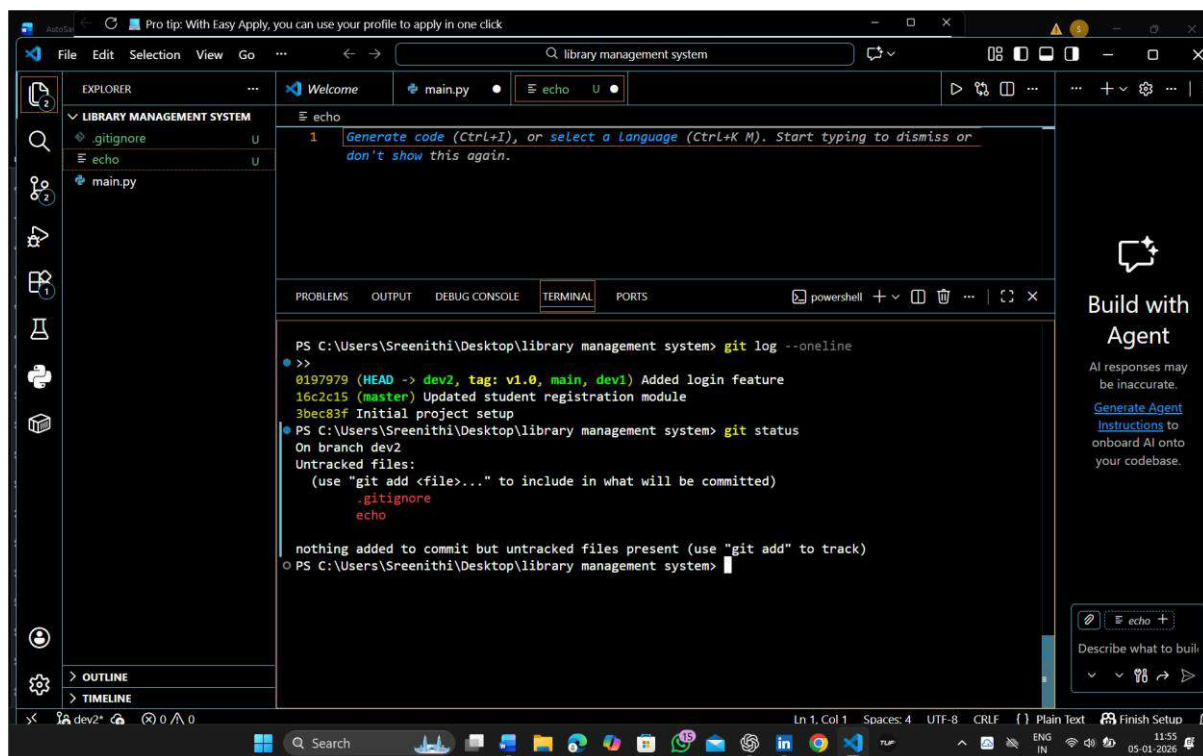
nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\Sreenithi\Desktop\library management system>
```

Case Study 10: Final Project Release

Scenario: Project is ready for final submission.

Tasks:

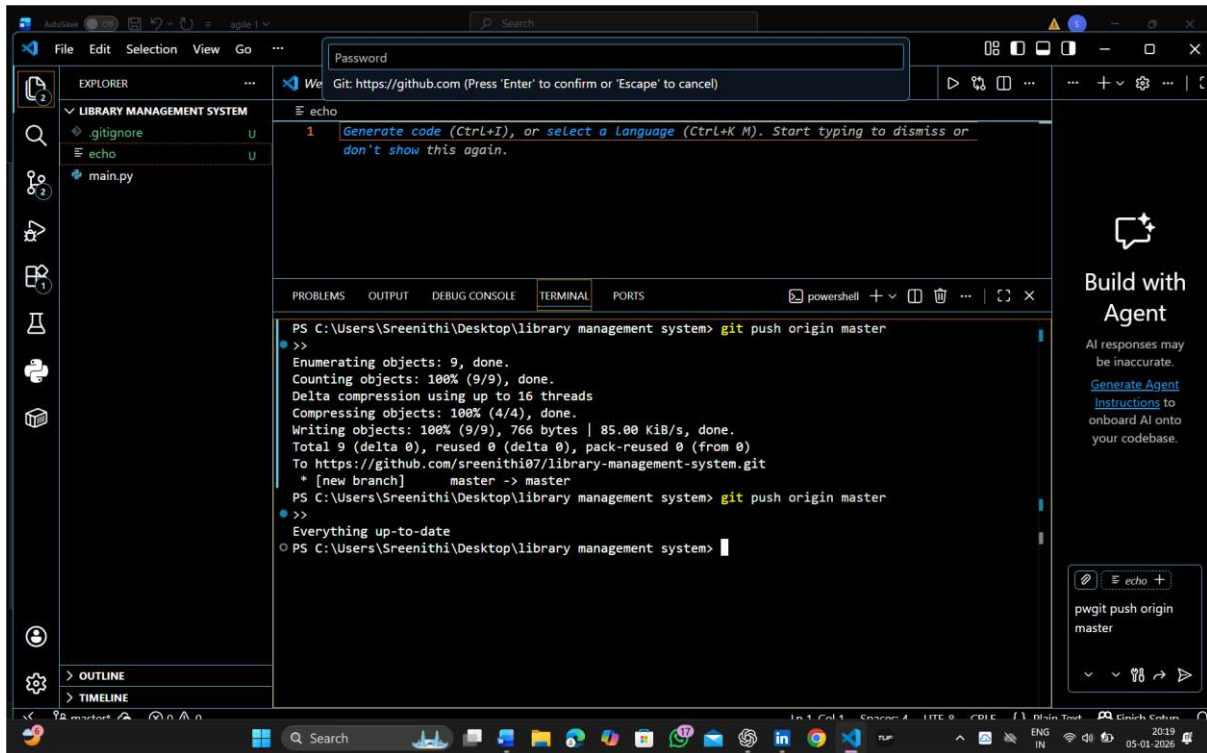
- Review commit history



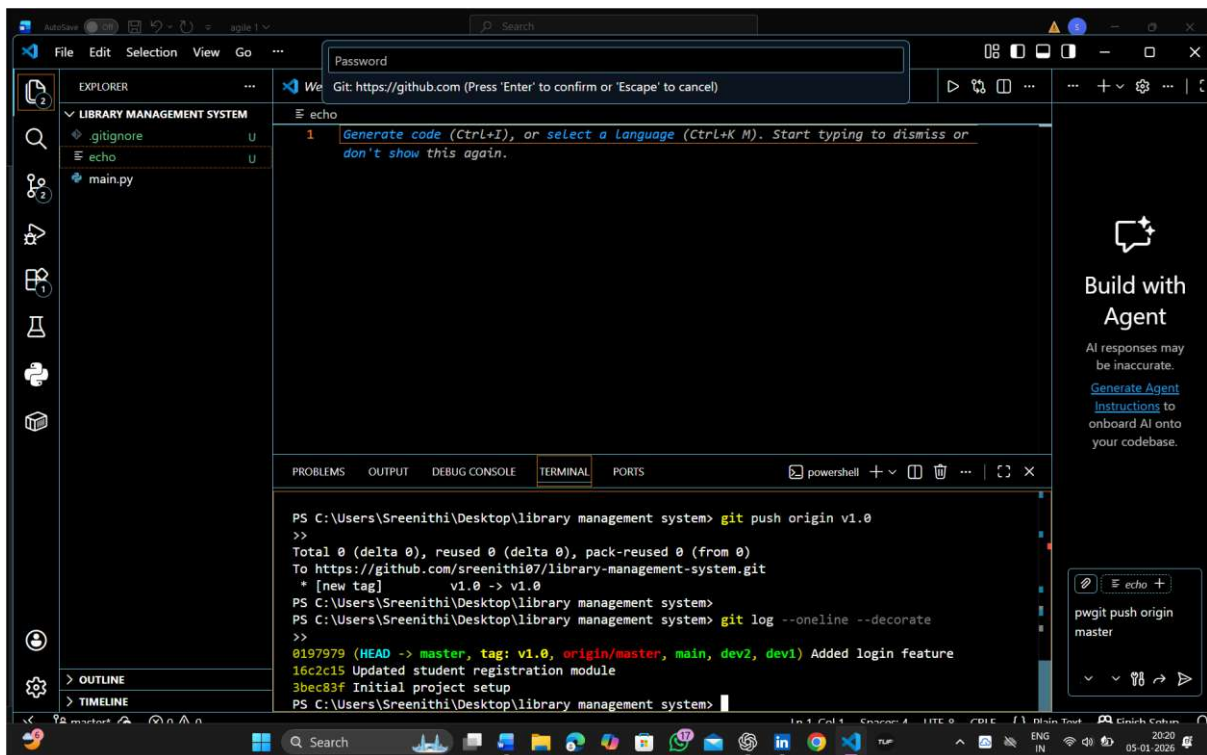
```
PS C:\Users\Sreenithi\Desktop\library management system> git log --oneline
>>
0197979 (HEAD -> dev2, tag: v1.0, main, dev1) Added login feature
16c2c15 (master) Updated student registration module
3bec83f Initial project setup
PS C:\Users\Sreenithi\Desktop\library management system> git status
On branch dev2
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        echo

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\Sreenithi\Desktop\library management system>
```

- Push final code



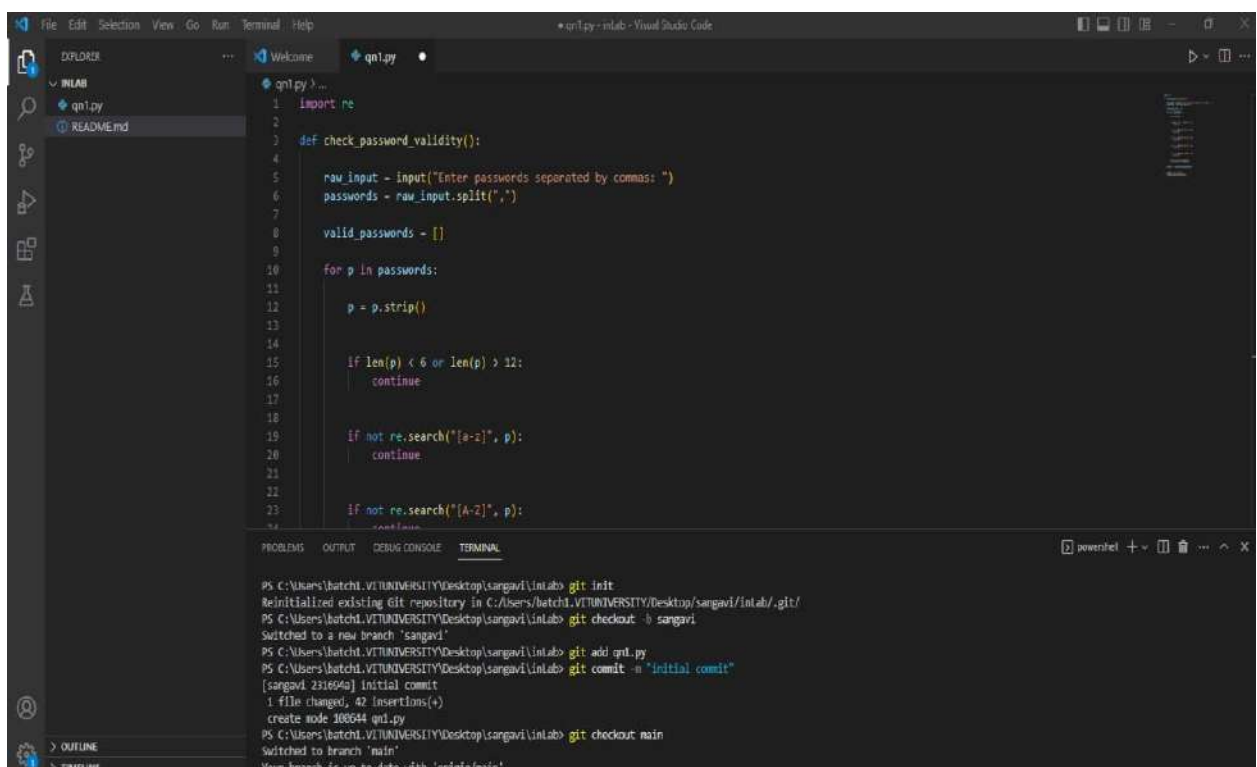
- Prepare release version



INLAB TASK:

Team collaboration - worked in different branches simultaneously and resolved conflicts and pushed the final version.

Qn1 Password checker:

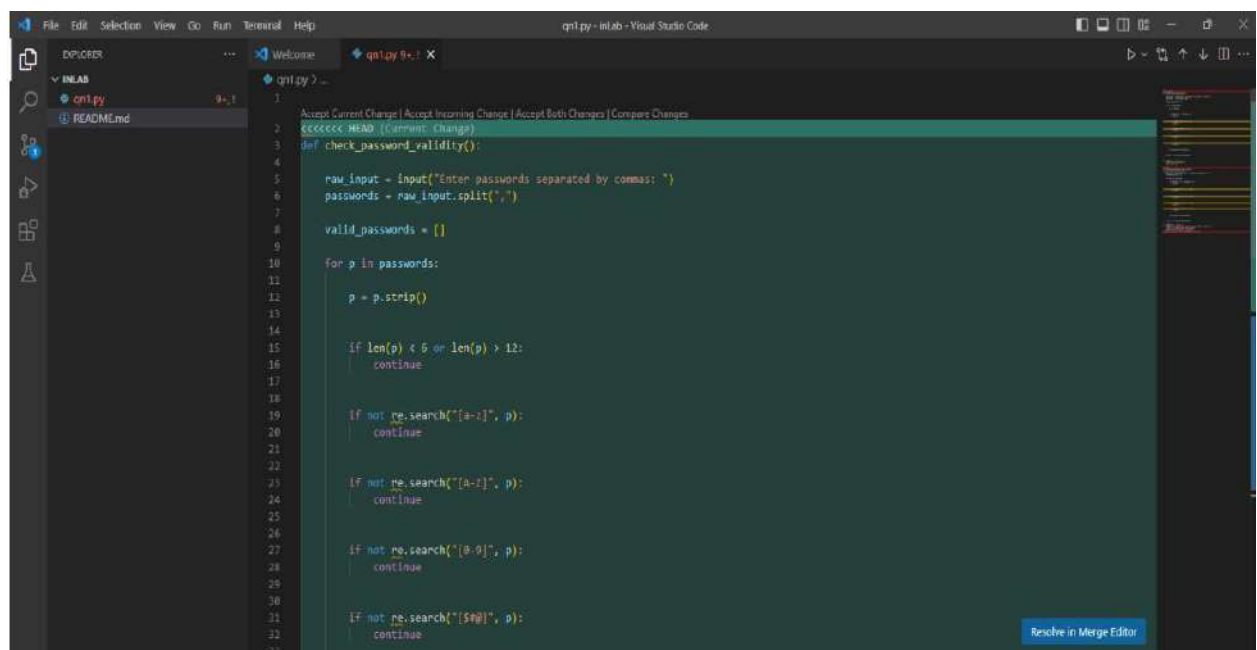


The screenshot shows the Visual Studio Code interface with a file explorer on the left containing 'INLAB', 'qn1.py', and 'README.md'. The main editor displays the 'qn1.py' file with the following code:

```
1 import re
2
3 def check_password_validity():
4
5     raw_input = input("Enter passwords separated by commas: ")
6     passwords = raw_input.split(",")
7
8     valid_passwords = []
9
10    for p in passwords:
11
12        p = p.strip()
13
14
15        if len(p) < 6 or len(p) > 12:
16            continue
17
18
19        if not re.search("[a-z]", p):
20            continue
21
22
23        if not re.search("[A-Z]", p):
24            continue
25
26    return valid_passwords
```

The bottom panel shows the terminal output of the following commands:

```
PS C:\Users\batch1.VTUNIVERSITY\Desktop\sangavi\inlab> git init
Reinitialized existing Git repository in C:\Users\batch1.VTUNIVERSITY\Desktop\sangavi\inlab\.git\
PS C:\Users\batch1.VTUNIVERSITY\Desktop\sangavi\inlab> git checkout -b sangavi
Switched to a new branch 'sangavi'
PS C:\Users\batch1.VTUNIVERSITY\Desktop\sangavi\inlab> git add qn1.py
PS C:\Users\batch1.VTUNIVERSITY\Desktop\sangavi\inlab> git commit -m "initial commit"
[sangavi 231694a] initial commit
1 file changed, 42 insertions(+)
create mode 100644 qn1.py
PS C:\Users\batch1.VTUNIVERSITY\Desktop\sangavi\inlab> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```



The screenshot shows the Visual Studio Code interface with a merge conflict resolution window open. The 'qn1.py' file is shown with a conflict between the 'main' branch and a branch named 'sangavi'. The conflict is resolved by accepting the changes from the 'sangavi' branch. The code in the editor is identical to the previous screenshot, but now includes a conflict resolution bar at the top of the editor area.

Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes

<<<<<<< HEAD (Current Change)

```
1 def check_password_validity():
2
3     raw_input = input("Enter passwords separated by commas: ")
4     passwords = raw_input.split(",")
5
6     valid_passwords = []
7
8     for p in passwords:
9
10        p = p.strip()
11
12
13        if len(p) < 6 or len(p) > 12:
14            continue
15
16
17        if not re.search("[a-z]", p):
18            continue
19
20
21        if not re.search("[A-Z]", p):
22            continue
23
24        if not re.search("[0-9]", p):
25            continue
26
27        if not re.search("[!@#]", p):
28            continue
29
30    return valid_passwords
```

Resolve in Merge Editor

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git checkout -b sangavi1
Switched to a new branch 'sangavi1'
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git add qn1.py
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git commit -m "modified the code"
[sangavi1 4168f13] modified the code
1 file changed, 10 insertions(+), 10 deletions(-)
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git checkout main
error: Your local changes to the following files would be overwritten by checkout:
    qn1.py
Please commit your changes or stash them before you switch branches.
Aborting
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git merge sangavi
Already up to date.
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git merge sangavi1
Already up to date.
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git add qn1.py
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git commit -m "initial commit"
[sangavi1 1e6a81a] initial commit
1 file changed, 3 insertions(+), 1 deletion(-)
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git checkout main
Switched to branch 'main'
Your branch is behind 'origin/main' by 1 commit, and can be fast-forwarded.
(use 'git pull' to update your local branch)
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git merge sangavi1
Updating 231694a..1e6a81a
Fast-forward
 qn1.py | 24 ++++++-----
1 file changed, 13 insertions(+), 11 deletions(-)
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git push origin main
to https://github.com/Abi503/inLab.git
! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/Abi503/inLab.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git pull origin main
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git pull origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 3 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 320 bytes | 21.00 KiB/s, done.
From https://github.com/Abi503/inLab
 * branch          main       -> FETCH_HEAD
  be9912a..1ccf34a  main       -> origin/main
Auto-merging qn1.py
Automatic merge failed; fix conflicts and then commit the result.
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git add qn1.py
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git commit -m "resolved conflict"
[main 7e7c669] resolved conflict
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git push origin main
fatal: unable to access 'https://github.com/Abi503/inLab.git/': Could not resolve host: github.com
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab> git push origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 20 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 811 bytes | 811.00 KiB/s, done.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/Abi503/inLab.git
 1ccf34a..7e7c669  main -> main
PS C:\Users\batch1.VITUNIVERSITY\Desktop\sangavi\inLab>
```

Team collaboration -one team member pushed the code the other pulled and modified and then pushed as final version

Qn5 Monthly salary:

The image shows a Visual Studio Code editor window with a Python file named `qn5.py` and a terminal window. The Python script calculates the monthly salary based on basic salary, HRA, DA, and tax. The terminal window shows the execution of git commands to initialize a repository, checkout a branch, add the file, and commit it.

```
1 # Accept basic salary
2 basic = float(input("Enter Basic Salary: "))
3
4 # Calculate HRA and DA
5 hra = 0.20 * basic
6 da = 0.10 * basic
7
8 # Calculate total (gross) salary
9 gross_salary = basic + hra + da
10
11 # Calculate tax
12 tax = 0.05 * gross_salary
13
14 # Calculate net salary
15 net_salary = gross_salary - tax
16
17 # Display results
18 print("Basic Salary:", basic)
19 print("HRA (20%):", hra)
20 print("DA (10%):", da)
21 print("Gross Salary:", gross_salary)
22 print("Tax (5%):", tax)
23 print("Net Salary:", net_salary)
24
```

```
PS C:\Users\radha\OneDrive\Desktop\inlab> git init
Initialized empty Git repository in C:\Users\radha\OneDrive\Desktop\inlab\.git\
PS C:\Users\radha\OneDrive\Desktop\inlab> git checkout -b abinaya
Switched to a new branch 'abinaya'
PS C:\Users\radha\OneDrive\Desktop\inlab> git add qn5.py
PS C:\Users\radha\OneDrive\Desktop\inlab> git commit -m "Initial commit"
[abinaya (root-commit) 790e48f] initial commit
1 file changed, 23 insertions(+)
create mode 100644 qn5.py
PS C:\Users\radha\OneDrive\Desktop\inlab>
```

The image shows a Visual Studio Code editor window with a Python file named `qn5.py` and a terminal window. The Python script calculates the monthly salary based on basic salary, HRA, DA, and tax. The terminal window shows the execution of git commands to initialize a repository, checkout a branch, add the file, and commit it.

```
1 # Accept basic salary
2 basic = float(input("Enter Basic Salary: "))
3
4 # Calculate HRA and DA
5 hra = 0.20 * basic
6 da = 0.10 * basic
7
8 # Calculate total (gross) salary
9 gross_salary = basic + hra + da
10
11 # Calculate tax
12 tax = 0.05 * gross_salary
13
14 # Calculate net salary
15 net_salary = gross_salary - tax
16
17 # Display results
18 print("Basic Salary:", basic)
19 print("HRA (20%):", hra)
20 print("DA (10%):", da)
21 print("Gross Salary:", gross_salary)
22 print("Tax (5%):", tax)
23 print("Net Salary:", net_salary)
24
```

```
PS C:\Users\radha\OneDrive\Desktop\inlab> git init
Initialized empty Git repository in C:\Users\radha\OneDrive\Desktop\inlab\.git\
PS C:\Users\radha\OneDrive\Desktop\inlab> git checkout -b abinaya
Switched to a new branch 'abinaya'
PS C:\Users\radha\OneDrive\Desktop\inlab> git add qn5.py
PS C:\Users\radha\OneDrive\Desktop\inlab> git commit -m "Initial commit"
[abinaya (root-commit) 790e48f] initial commit
1 file changed, 23 insertions(+)
create mode 100644 qn5.py
PS C:\Users\radha\OneDrive\Desktop\inlab>
```

